VF Asia Footwear Laboratory Testing Manual

Test Method for Crush Strength	Test method # VF- B12 Page 1 of 2
Date Issued: May,2019 **Revision #: 1** **Reason for revision: Rectify the unit of measure from kg to kgf for material strength.**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the ability of the material to resist collapsing as a result of being compressed between two flat horizontal plates.

REFERENCES: None

EQUIPMENT AND MATERIAL:

1. Tensile testing machine, meeting the requirements of ASTM D638
2. Compression platens , top plate has smooth surface , bottom plate has a circular groove cut centrally in it , as follows:

   5.0 ± 0.2 mm deep
   2.5 ± 0.2 mm wide
   49.0 ± 0.2 mm, outside diameter of the circular groove
   44.0 ± 0.2 mm, inside diameter of the circular groove

3. Heel counter, paperboard / box

SAMPLE PREPARATION:

1. Condition the sample for 24 hours at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

2. Die cut the sample to a width of 2.54 +/- 0.1 cm and a length of 15.0+/- 0.1 cm. Samples should always be cut such that the width direction corresponds to the height of the heel counter or paperboard/box, and such that the sample is taken from near the base of the heel counter.

REPETITIONS: Test three (3) samples per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Measure the thickness at three points along the sample and record the average to the nearest 0.01 mm

2. Equip the tensile testing machine with the compression plates

Test Method for Crush Strength	Test method # VF- B12 Page 2 of 2
Date Issued: May,2019 Revision #: 1 Reason for revision: Rectify the unit of measure from kg to kgf for material strength.	Created by: Nila Babia Erazo

3. Zero the tensile testing machine , Set the crosshead speed to 25 mm/min
4. Position the test sample in the groove, if testing sample is too long, trim the length until the ends of the test sample butt when inserted in the groove.

Note: If the sample is over-trimmed, it must be discarded. It is not acceptable to have any gap between the ends of the test samples. Also, tape is not to be used to hold the two ends of the test sample together.

5. Lower the top compression plate onto the test sample and continue to compress it until the sample collapses.
6. Record the maximum load to the nearest 0.1 kgf

REPORT: The average of the three samples to the nearest 0.1kgf.

Note: The thickness of the test sample is not factored into a crush strength measurement. However, it is important to measure the thickness of the test sample to verify the thickness for reference purpose.